



भारतीय मानक ब्यूरो

(उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय, भारत सरकार)

BUREAU OF INDIAN STANDARDS

(Ministry of Consumer Affairs, Food & Public Distribution, Govt. of India)

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व्यापक परिचालन मसौदा

हमारा संदर्भ : सीईडी 06/टी-6

17 अगस्त 2026

तकनीकी समिति : पत्थर विषय समिति, सीईडी - 06

प्राप्तकर्ता :

- क) सिविल इंजीनियरी विभाग परिषद्, सीईडीसी के सभी सदस्य
ख) सीईडी 06 के सभी सदस्य
ग) रूचि रखने वाले अन्य निकाय

प्रिय महोदय/महोदया,

निम्नलिखित भारतीय मानक का मसौदा संलग्न हैं:

प्रलेख संख्या	शीर्षक
सीईडी 06 (35146)WC	भारतीय मानक का मसौदा प्राकृतिक पत्थरों की पहचान के तरीके (IS 3316 का दूसरा पुनरीक्षण) ICS No. 91.100.15

कृपया इस मानक के मसौदे का अवलोकन करें और अपनी सम्मतियाँ यह बताते हुए भेजे कि यदि यह मानक के रूप में प्रकाशित हो तो इस पर अमल करने में आपके व्यवसाय अथवा कारोबार में क्या कठिनाइयाँ आ सकती हैं।

सम्मतियाँ भेजने की अंतिम तिथि: 01 अक्टूबर 2026

टिप्पणियाँ, यदि कोई हों, बीआईएस ई-गवर्नेंस पोर्टल के https://www.services.bis.gov.in/php/BIS_2.0/dgdashboard/draft/darftdetail/63/3/CED के माध्यम से ऑनलाइन भेजी जा सकती हैं।

वैकल्पिक रूप से, टिप्पणियाँ संलग्न प्रारूप में भी दर्ज की जा सकती हैं और ced6@bis.gov.in या ced@bis.gov.in पर ईमेल की जा सकती हैं।

आपको अपनी टिप्पणियाँ प्रस्तुत करने के लिए लॉगिन करना पड़ सकता है, कृपया लॉगिन बनाएं।

यदि कोई सम्मति प्राप्त नहीं होती है अथवा सम्मति में केवल भाषा सम्बन्धी त्रुटि हुई तो उपरोक्त प्रलेख को यथावत अंतिम रूप दिया जाएगा। यदि सम्मित तकनीकी प्रकृति की हुई विषय समिति के अध्यक्ष के परामर्श से अथवा उनकी इच्छा पर आगे की कार्यवाही के लिए विषय समिति को भेजे जाने के बाद प्रलेख को अंतिम रूप दे दिया जाएगा।

यह प्रलेख भारतीय मानक ब्यूरो की वेबसाइट www.bis.gov.in पर भी उपलब्ध हैं।

धन्यवाद।

भवदीय

ह/-

(श्री द्वैपायन भद्र)

वैज्ञानिक 'ई'/निदेशक एवं प्रमुख (सिविल इंजीनियरिंग)

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संलग्न: उपरलिखित



भारतीय मानक ब्यूरो
(उपभोक्ता मामले, खाद्य एवं सार्वजनिक वितरण मंत्रालय, भारत सरकार)
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WIDE CIRCULATION DRAFT

Our Reference: CED 06/T-6

17 Augst 2026

Technical Committee: Stone Sectional Committee, CED 06

Addressed To:

- All Members of Civil Engineering Division Council, CEDC
- All Members of CED 06
- All others interested

Dear Sir/Madam,

Please find enclosed the following document:

Doc No.	Title
CED 06 (35146)WC	Draft Indian Standard Methods of Identification of Natural Stones (second revision of IS 1123) ICS No. 91.100.15

Kindly examine the draft standard and forward your views stating any difficulties which you are likely to experience in your business or profession, if this is finally adopted as National Standard.

Last Date for comments: 01 October 2026

Comments if any, may be sent online through the BIS e-governance portal at https://www.services.bis.gov.in/php/BIS_2.0/dgdashboard/draft/darftdetail/63/3/CED .

Alternatively, comments may also be recorded in the enclosed format and emailed at ced6@bis.gov.in or at ced@bis.gov.in.

You may be required to login to submit your comments, kindly create a login.

In case no comments are received or comments received are of editorial nature, you will kindly permit us to presume your approval for the above document as finalized. However, in case of comments of technical in nature are received then it may be finalized either in consultation with the Chairman, Sectional Committee or referred to the Sectional Committee for further necessary action if so desired by the Chairman, Sectional Committee.

The document is also hosted on BIS website www.bis.gov.in.

Thanking you,

Sd/-

(Shri Dwaipayan Bhadra)

Scientist 'E'/Director And Head
(Civil Engineering)

E-mail: hced@bis.gov.in

Encl: As above

FORMAT FOR SENDING COMMENTS ON THE DOCUMENT

[Please use A4 size sheet of paper only and type within fields indicated. Comments on each clause/sub-clause/table/figure, etc, be stated on a fresh row. Information/comments should include reasons for comments, technical references and suggestions for modified wordings of the clause. **Comments through https://www.services.bis.gov.in/php/BIS_2.0/WCDraft/comment_pdraft.php shall be appreciated.**]

Doc. No.: CED 06(35146) WC**BIS Letter Ref:** CED 06/T- 6**Title:** Methods of Identification of Natural Stones (*second revision* of IS 1123)Last date of comments: **01 October 2026****Name of the Commentator/ Organization:** _____

SI No.	Clause/ Para/ Table/ Figure No. commented	Comments/ Modified Wordings	Justification of Proposed Change
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NOTE- Kindly insert more rows as necessary for each clause/table, etc

Indian Standard

METHODS OF IDENTIFICATION OF NATURAL BUILDING STONES

(Second Revision)

FOREWORD

(Formal clause will be added later.)

Building stones are available in large quantities in various parts of the country. Their satisfactory selection and utilization require identification of the type of rock and determination of its relevant properties in accordance with standardized methods. Since the mineral constituents of a rock form the basis for its classification and identification, this standard provides methods for the identification of natural building stones. This standard was first published in 1957 and included provisions relating to the petrographical examination of building stones. In the 1975 revision, the scope has been limited to the identification of natural building stones, as required by research laboratories and organizations engaged in the examination and use of building stones. In this revision, the provisions have been editorially updated and references to relevant Indian Standards have been revised accordingly.

This standard contributes to the United Nations Sustainable Development Goal 9 'Industry, innovation and infrastructure'; and Goal 12 'Ensure sustainable consumption and production patterns' towards substantial reduction of waste generation through prevention, reduction, recycling and reuse.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a measurement shall be rounded off in accordance with IS 2 : 2022 'Rules for rounding off numerical values (second revision)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

METHODS OF IDENTIFICATION OF NATURAL BUILDING STONES

(Second Revision)

1 SCOPE

1.1 This standard lays down the procedure for identification of some of common types of natural building stones.

2 SELECTION OF SAMPLE

2.1 The sample shall be selected so as to represent the type and grade of the stone under consideration. The sample shall be obtained from fresh, unweathered rock.

2.2 The sample shall be selected by the purchaser or the purchaser's authorised representative from the quarried stone or obtained from the natural rock, as described specified in **2.2.1** and **2.2.2** and shall be of adequate size to permit the preparation of the required number of test specimens.

2.2.1 *Stones from Ledges or Quarries* – The ledge or quarry face of the stone shall be inspected to determine any variation between the different strata. Differences in colour, texture, and structure shall be observed. Separate samples of stone weighing at least 25 kg each weighing not less than 25 kg shall be obtained from all strata that appear to vary in colour, texture and structure. Pieces that have been damaged by blasting, driving wedges, heating or other similar operations, shall not be included in the sample.

2.2.2 *Field Stones and Boulders* – A detailed inspection of the stone and boulders over the area from which the supply is proposed to be obtained shall be made. The different kinds types of stone and their condition at various quarry sites shall be recorded. Separate samples for each class of stone that would be considered for use in construction as indicated by visual inspection shall be selected.

3 PROCEDURE

3.1 The sample shall be examined macroscopically for its colour, structure, texture, and mineral constituents.

3.2 The type of rock shall be identified in accordance with characteristics given in Table 1. In case of doubt guidance can be obtained from engineering properties of the rock given in Table 1.

4 REPORTING

4.1 Date of sample taken, identification of sample and the type of stone shall be reported.

TABLE 1 CHARACTERISTICS OF BUILDING STONES

(Clause 3.2)

SI No.	TYPE	PHYSICAL PROPERTIES			USES	AVAILABILITY	AVERAGE ENGINEERING PROPERTIES (See NOTE)						
		Colour	Texture and structure	Mineralogical Composition			Specific Gravity	Compressive Strength	Structure Strength	Tensile Strength	Porosity	Resistance to Abrasion	Modulus of Elasticity
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
								kg/cm ²	kg/cm ²	kg/cm ²	Percent	Percent	kg/cm ²
Class: Igneous rocks													
1.	Granite*	White to light grey and pink.	Crystalline, fine to coarse grained, massive sometimes sheeted and banded; joints common.	Essentially quartz and feldspar with mica, amphiboles and pyroxenes as accessories.	Used primarily for bridge piers, river walls, dams and related structure, pavements, kerbs, pedestral, monumental buildings, table, top, coarse aggregate, road metal, etc. polished granite looks more elegant than marble with a very long lasting lustre. Good foreign market.	Granites occur throughout the country. Only relatively small number of suitable colour or texture are available. Most granites are characterised by joints and fractures. It is desired that granites shall be free from Ram, etc. Granites may be graded by their compressive strength.	2.63 to 2.75	1 000 to 2 500	140 to 500	70 to 250	0.40 to 4	43.9 to 87.9	2 × 10 ⁵ to 6 × 10 ⁵
2.	Granodiorite*	Light grey	Crystalline, medium to coarse grained; massive; joints common.	Essential minerals are quartz and plagioclase feldspar with accessories like biotite and hornblende.	Same use as granites.	Less abundant in occurrence.	2.80 to 3.00	—	—	—	0.50	—	—
3.	Syenite*	Light coloured to dark green, grey and bluish grey.	Crystalline, medium to coarse grained; massive; joints common.	Essential minerals are alkali felspar sometimes with nepheline (nepheline syenite). Common accessories are hornblende, biotite and augite.	Same use as granites.	Syenites are less abundant than granites.	2.60 to 2.80	350 to 500	—	—	1.38 to 1.54	—	6 × 10 ⁵ to 8 × 10 ⁵
4.	Diorite*	Grey to dark grey.	Crystalline, medium to coarse grained; massive; joints common.	Essential minerals are plagioclase feldspar and dark minerals.	Used as a good aggregate material and road metal, etc.	Found in a number of places in India.	2.80 to 3.00	1 800 to 3 000	—	150 to 300	0.25	—	7 × 10 ⁵ to 10 × 10 ⁵

5.	Gabbro*	Dark grey to black	Crystalline, medium to coarse grained; banded and often jointed.	Consists of lime-feldspar and pyroxene (augite) accessories may be olivine, biotite, hornblende and rarely quartz.	May be used where available for bridge piers, river walls dam and related structures; it may also be used for pavements, kerbs and in buildings (same as granite).	It's high strength value and low porosity, makes it suitable for heavy structures.	2.90 to 3.20	1 800 to 3 000	–	150 to 300	0.10 to 0.20	–	7 × 10 ⁵ to 11 × 10 ⁵
6.	Basalt (Deccan trap)**	Dark grey to black	Medium to fine grained, dense and compact and also at times vesicular with flow bands. Often occurring in columns and prismatic structures. Fractures and joints common.	Plagioclase feldspar and pyroxene with or without interstitial glass; olivine, sometimes common. Vesicles filled by quartz, calcite and zeolite.	Used primarily for bridge piers, river walls, dams and related structures, in masonry works and is used also for pavements, kerbs, monumental buildings; etc.	Basalts occur as “trap” rocks in western India, and in Rajmahal. It is cut into slabs and blocks. The presence of void, in filling of calcite, gcolite and quartz are undesirable. Fractures, joints also decrease its importance. Basalt generally have a dark brown have a dark brown appearance.	2.60 to 3.00	1 500 to 3 000	200 to 600	100 to 300	0.10 to 1.00	14.86 to 18.92	6 × 10 ⁵ to 10 × 10 ⁵
7.	Dolerite**	Dark grey to black	Crystalline, fine to coarse grained; joints and fractures common.	Consists of plagioclase feldspar and pyroxene, olivine, hornblende and biotite are common accessories.	Same as gabbro.	It's high strength values and low porosity makes it suitable for heavy structures. But joints may not allow large homogeneous block and slabs to be cut from quarries. Makes a very high quality aggregate material.	3.00 to 3.05	2 000 to 3 500	250 to 600	150 to 350	0.10 to 0.50	–	8 × 10 ⁵ to 11 × 10 ⁵
8.	Rhyolite**	White to light, grey, pink and greyish black; black when extremely glassy (obsidian and pitchstone).	Glassy to cryptocrystalline with few embedded crystals of quartz and feldspars; flowbands, and fractures quite common.	A volcanic equivalent of granite.	Though not used on large scale; rhyolite may be used as blocks if easily available; in buildings, paving blocks, ornamental and decorative works, etc.	Rhyolites are less abundant than granites. The rock may be characterised by cracks and fractures which render the blocks unsuitable for the purpose of building stone.	2.40 to 2.60	–	–	–	4 to 6	–	–
9.	Trachyte**	Grey to bluish grey.	Glassy, aphanitic; flow bands and fractures common.	A volcanic equivalent of syenite.	Used as a building stone – dimensions stone, grinding stone (same as granite).	Fairly abundant in occurrence.	2.60 to 2.85	820	–	–	–	19.50	–

10.	Andesite **	Dark grey to black.	Glassy, aphanitic.	A volcanic rock with plagioclase feldspar and one or more minerals like biotite pyroxene and hornblende.	Not generally in use, but may be used, if easily available, as blocks and slabs in buildings paving blocks ornamental and decorative works, etc.	It is not abundant in India. May possesses cooling cracks, joints and fractures which may render it unsuitable as a building stone.	2.20 to 2.60	1 300 to 2 500	—	—	0.10 to 11	—	—
Class: Sedimentary Rock													
1.	Sandstone	Depends on the colour of matrix and cement; generally white, grey, red, buff, brown, yellow and even dark grey.	Stratified (bedded) fine to coarse grained; cross-bedding flute-cast and sole markings, etc, common.	Consists mainly of quartz with feldspar and dark minerals in a siliceous argillaceous and feeruginous cement.	Used for masonry work, for dams, bridge piers, river, walls, buildings, pavements, kerbs, monumental stones, etc.	Famous building stone, ornamental stone. Weathering of the rock renders it unsuitable as building stone. Kaolinization of feldspars and leaching effect of matrix and cement make the rock porous. It is desirable to use silica cement for heavy structures if necessary.	1.85 to 2.70	200 to 1 700	80 to 400	40 to 250	5 to 25	1.60 to 29.00	0.50 × 10 ⁵ to 8.00 × 10 ⁵
2.	Limestone and dolomite	White, grey, pink, red, blue, buff brown, green yellow, black, etc. (colour due to impurities in the form of silicates and carbonates).	Bedded, granular; fine grained.	Consists essentially of calcite (calcium carbonate) with varying amounts of magnesium carbonate.	Generally used as slabs and tiles in any type of construction used in buildings. Also large size blocks of Milliolite limestone (Porbandar stone) used as ornamental and building stone.	Limestone is found to occur throughout India. It should be devoid of any argillaceous band, softer vein, cracks	2.14 to 2.80 (limestone) 2.50 to 2.80 (dolomite)	300 to 2 500 (limestone) and (dolomite)	100 to 500	50 to 250	5 to 20	1.30 to 24.10	1.00 × 10 ⁵ to 8.00 × 10 ⁵
3.	Laterite	Brownish red, yellow, brown, grey and mottled colours.	Porous oolitic and pisolitic with cavities: at times bedded.	A mixture of hydrated oxides of iron and aluminium frequently with manganese dioxide, titanium oxide and free silica.	Generally used as blocks in the construction of buildings, institutional and commercial buildings.	Freshly quarried laterite is soft and porous, but when exposed to atmospheric conditions it hardens and makes a very tough material. When used in walls it should be plastered from outside.	1.85	19 to 23	—	—	—	—	—

Class: Metamorphic Rocks

1.	Charnockite	Light grey to dark grey.	Fine to coarse grained, massive banded and sometimes foliated.	Consists of quartz, feldspar, hypersthene and garnet.	Generally used as slabs and block in the construction of buildings, monuments, pavements, kerbs, etc, form a source of aggregate material.	Occurs in association with Khondalites in the Eastern Ghats. Its strength is similar to granite and may be put to similar uses as granite.	2.70 to 3.00	—	—	—	—	—	7.94 × 10 ⁵ to 9.94 × 10 ⁵
2.	Gneiss	Light grey, pink, purple, greenish grey and dark grey.	Fine to coarse grained; alternative dark and white bands (gneissose structure).	Composed of quartz feldspar, biotite, hornblends, etc.	Not commonly used because of deleterious constitutes, but may be used in minor constructions if easily available. Hard gneisses may be used for construction of buildings and decorative works and also as riprap stone.	Varying composition of bands are likely to give low strength of the rocks. Widely found in India.	2.50 to 3.00	500 to 2 500	—	50 to 200	0.50 to 1.50	—	2.01 × 10 ⁵ to 4.00 × 10 ⁵
3.	Quartzite	White, grey, yellowish & brownish grey, buff (colour is dependent on the impurities in the cement).	Fine to coarse grained often granular and banded.	Quartz is the chief constituent with feldspars and mica in small amounts.	Used as blocks and slabs for buildings stone and aggregate material.	Widely available in India.	2.55 to 2.65	1 500 to 3 000	200 to 600	100 to 300	0.20 to 0.60	—	9.3 × 10
4.	Marble	White, rose, pink, red, green, yellow, black, etc. (impurities impact different shades).	Fine to coarse grained, massive crystalline and granular.	Mainly calcite and dolomite or mixture of the two with some other impurities, e.g. wallastonite and serpentine as bands and veins.	Used as blocks, slabs and tiles in monuments, temples, commercial buildings. Coloured marbles are as ornamental stones.	Marbles may contain serpentine band, valuable for decorative works.	2.60 to 2.70	1 000 to 2 500	150 to 300	70 to 200	2 to 4	6.70 to 41.70	-
5.	Khondalite	Light grey to brown and light pink.	Medium to fine grained; often banded or schistose.	Composed of quartz feldspar, garnet and sillimanite, with biotite and muscovite.	Khondalite is widely used for building stones in commercial and institutional buildings in the form	Khondalite may contain softer schistose bands and are found to weather easily on exposure to atmospheric condition. It	2.36 to 2.51	—	—	—	—	—	—

of blocks and slabs. In ancient times it was used in monuments and temples.

is easily available all along the Eastern Ghats.

6.	Slate	Dark grey, black, greenish grey, etc.	Fine grained, slaty cleavage, fissile along planes of original bedding.	Composed of quartz mica and clay minerals.	Used as slabs and roofing tiles in the construction of buildings, pavements, etc.	Easily cut into slabs, of desired dimensions. It may contain softer bands. Slate is found in many parts of the country.	2.60 to 2.70	1 000 to 2 000	–	70 to 200	0.10 to 1.00	–	–
7.	Phyllite	Grey, greenish, buff, reddish. Rock characterised by a shining silky lusture (phyllitic shean).	Fine grained, often characterised by foliation, cleavage and bedding.	Composed of micas (commonly sericite), quartz, garnet and alusite, clay minerals and iron oxide.	Used as slabs roofing tiles for minor construction works and also caring statues in temples.	The rock is fissile. Harder phyllites (almost nearing to slates) are used commonly where their occurrence is abundant.	2.60 to 2.80	–	–	–	–	–	–
8.	Schists	Grey to black, brown, purple and greenish grey.	Fine to coarse grained, schistose.	Consists of quartz, feldspar, biotite, muscovite, chlorite, hornblende, garnet, etc.	Not generally used, because of its low strength, but if easily available may be used for minor construction. There are different types of schists, quartz-muscovite-quartz-sericite schists, hornblendes-schists, etc.	Do	2.31 to 3.04	400 to 950	–	–	–	–	1.80 × 10 ⁵ to 3.40 × 10 ⁵

NOTE – These values are only for guidance. The exact data is being collected and these values will be revised.

*These fall under 'Granite Group'.

**These fall under 'Basalt Group'.